



AI Evaluation Report

Data Science Expert — Self-Assessment

CANDIDATE

Mary Rod

FINAL SCORE

18 / 70

weighted points

CORRECT ANSWERS

9 / 35

questions

NEEDS IMPROVEMENT

Assessment Date: August 09, 2026 Report Generated: August 09, 2026 at 10:35 UTC Role: Data Science Expert

OVERALL RATING

★☆☆☆☆

SUCCESS RATE

25.7%

WEIGHTED SCORE

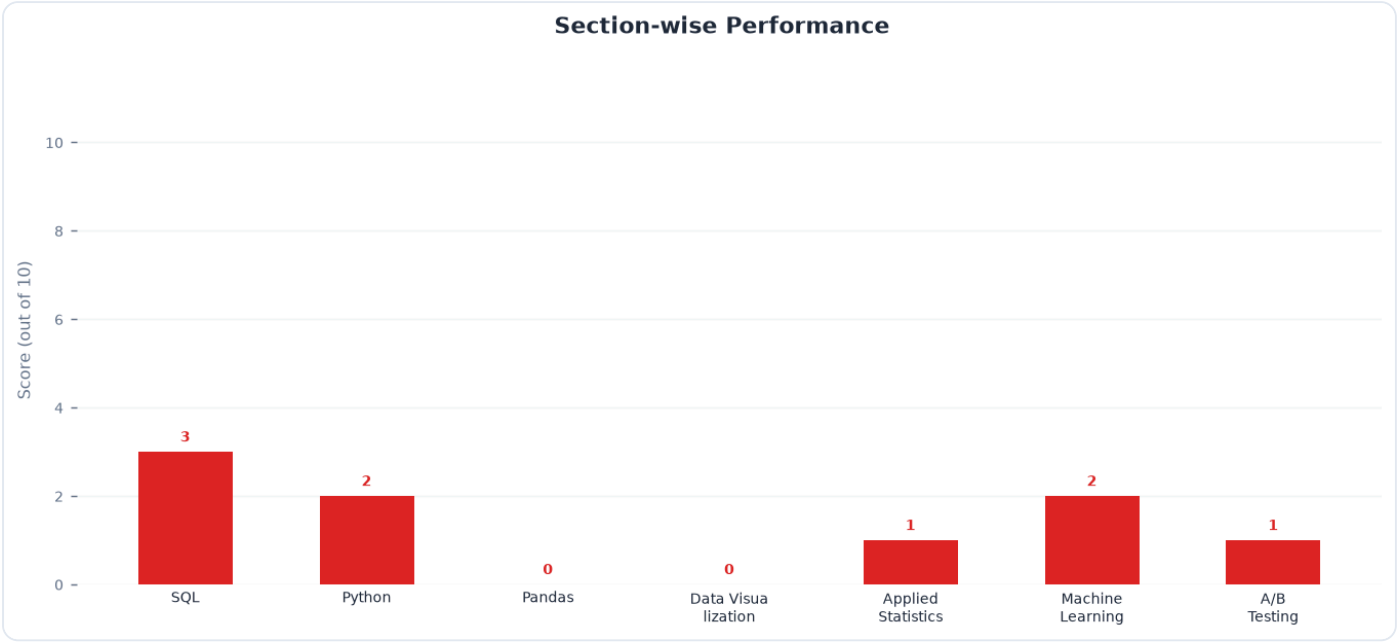
18 / 70

TOP SKILL

SQL

NEEDS IMMEDIATE ATTENTION

Pandas



Skill Rating & Benchmark

Section	Your Score	Reference Benchmark (Demo)	Delta	Status
SQL	3 / 10	6.8	-3.8	Needs Work
Python	2 / 10	6.2	-4.2	Critical Gap
Pandas	0 / 10	5.8	-5.8	Critical Gap
Data Visualization	0 / 10	6.0	-6.0	Critical Gap
Applied Statistics	1 / 10	5.5	-4.5	Critical Gap
Machine Learning	2 / 10	5.2	-3.2	Needs Work
A/B Testing	1 / 10	6.4	-5.4	Critical Gap

Attempt Statistics Breakdown

Section	Total Q	Attempted	Correct	Incorrect	Skipped	Accuracy
SQL	5	5	3	2	0	60%
Python	5	5	2	3	0	40%
Pandas	5	5	0	5	0	0%
Data Visualization	5	5	0	5	0	0%
Applied Statistics	5	5	1	4	0	20%
Machine Learning	5	5	2	3	0	40%
A/B Testing	5	5	1	4	0	20%

Executive Summary

The candidate achieved a score of 9 out of 35 correct answers (25.7% accuracy) with a weighted score of 18 out of 70 on this technical assessment. While the results indicate exposure to advanced data science concepts, significant gaps in core execution, syntax, and foundational statistical theory must be addressed. This evaluation outlines key areas for improvement to elevate the candidate's technical proficiency to a professional standard.

Key Strengths

- Conceptual Awareness:** Familiarity with high-level industry methodologies such as CUPED in A/B testing and ensemble model architectures.
- Basic SQL Structure:** Understanding of standard aggregation patterns, table grouping, and basic conditional logic.
- Python Flow Control:** Ability to read and interpret basic function definitions and exception handling blocks.

Top Blind Spots

- **SQL Window Functions & Indexing:** Misunderstanding how `ROW_NUMBER()` handles non-unique timestamps compared to correlated subqueries, and how composite indexes optimize these operations.
- **Pandas Execution & Slicing:** Confusion regarding label-based slicing inclusivity in `.loc` and the behavior of asynchronous joins using `pd.merge_asof`.
- **Applied Statistics & A/B Testing:** Critical gaps in fundamental probability (Bayes' Theorem calculations), distribution rules (empirical 68-95-99.7 rule), and experiment validation (Sample Ratio Mismatch mechanics).

Actionable Learning Resources

SQL Optimization: Complete the "Advanced SQL" track on Kaggle Learn, focusing specifically on window functions, partition clauses, and query execution plans.

✓ Action Item

Pandas Mastery: Work through the "Time Series" and "Merge, join, concatenate" sections of the official Pandas documentation, practicing `merge_asof` with synthetic financial datasets.

✓ Action Item

Statistical Foundations: Read Chapter 3 (Linear Regression) of *An Introduction to Statistical Learning (ISLR)*, focusing on residual diagnostics, heteroscedasticity, and polynomial feature engineering.

✓ Action Item

A/B Testing Practice: Study the "CUPED" and "Sample Ratio Mismatch" chapters in *Trustworthy Online Controlled Experiments* by Kohavi et al., to master variance reduction and experiment validation.

✓ Action Item